

Responsible AI Governance Framework

NovaMind AI Solutions

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Indiana Wesleyan University | 2025

Establishing positions, processes, and policies to oversee ethical AI design, development, and deployment across the enterprise.

Top Seven Responsible AI Job Roles

NovaMind's RAI governance structure assigns dedicated roles to ensure ethical accountability is embedded early and continuously throughout the AI lifecycle. These roles are informed by the NIST AI Risk Management Framework's emphasis on organizational governance and accountability, as well as guidance from *Ethical Machines* (Chapter 6) on building effective AI ethics organizations.

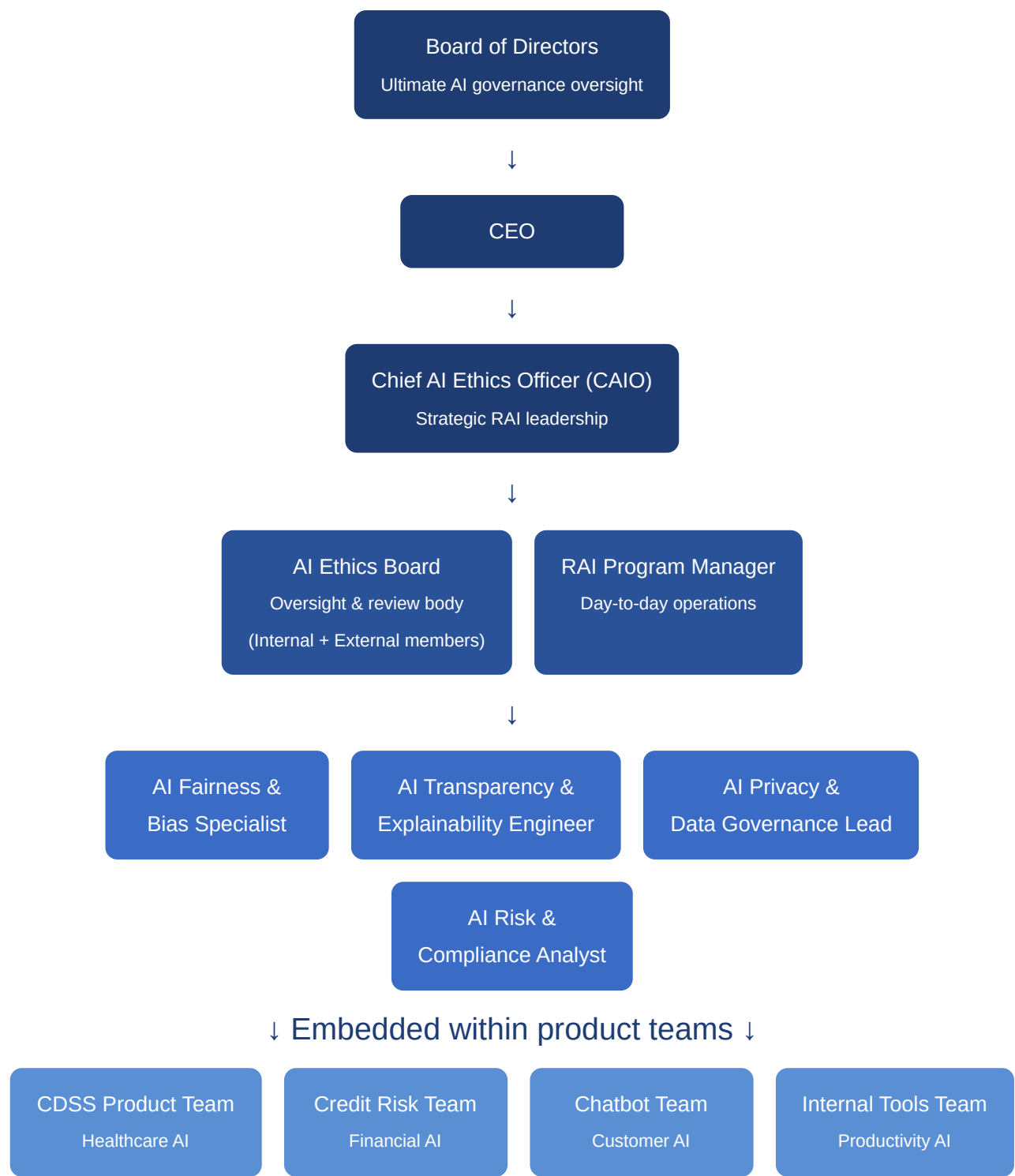
Role	Primary Responsibilities	Required Skills & Qualifications	Reports To
1. Chief AI Ethics Officer (CAIO)	Sets the strategic vision for responsible AI across the company. Chairs the AI Ethics Board. Represents AI ethics at the executive and board level. Makes final escalation decisions on high-risk AI deployments. Liaises with regulators and industry bodies on AI governance standards.	10+ years in technology leadership; deep knowledge of AI regulation (EU AI Act, NIST AI RMF, FDA AI guidance); background in ethics, law, or public policy; strong executive communication; experience managing cross-functional governance programs	CEO / Board of Directors
2. RAI Program Manager	Operationalizes the RAI framework day-to-day. Manages the RAI review pipeline and schedules Ethics Board reviews. Tracks compliance milestones, maintains documentation standards, coordinates training programs, and produces quarterly RAI status reports for leadership.	5+ years program/project management; PMP or equivalent certification; familiarity with AI/ML development lifecycle; risk management experience; strong organizational and documentation skills; stakeholder management across technical and non-technical teams	Chief AI Ethics Officer

Role	Primary Responsibilities	Required Skills & Qualifications	Reports To
3. AI Fairness & Bias Specialist	Designs and executes bias testing protocols for all AI models. Conducts disparate impact analyses. Recommends mitigation strategies for identified biases. Validates fairness metrics (equalized odds, demographic parity, calibration). Works with data teams to identify and address training data representativeness issues.	MS/PhD in data science, statistics, or related field; expertise in fairness metrics and bias detection tools (AI Fairness 360, Fairlearn); statistical testing proficiency; experience with protected class analysis; understanding of fair lending and anti-discrimination regulations	RAI Program Manager
4. AI Transparency & Explainability Engineer	Implements explainability solutions for AI products (SHAP, LIME, counterfactual explanations). Designs user-facing explanation interfaces. Validates that explanations accurately reflect model behavior. Develops explainability testing frameworks and integrates them into CI/CD pipelines.	MS in computer science or ML engineering; hands-on experience with XAI tools and libraries; strong software engineering skills; UX sensibility for designing comprehensible explanations; knowledge of model interpretability research; experience with both traditional ML and LLM explainability	RAI Program Manager
5. AI Privacy & Data Governance Lead	Ensures all AI data practices comply with HIPAA, GDPR, CCPA, and FCRA. Conducts privacy impact assessments for AI systems. Manages data	CIPP/US or CIPM certification; 5+ years in data privacy/governance; deep knowledge of healthcare and financial data regulations; experience with privacy-	Chief AI Ethics Officer (dotted line to General Counsel)

Role	Primary Responsibilities	Required Skills & Qualifications	Reports To
	retention and deletion policies. Oversees anonymization and de-identification processes. Reviews data sharing agreements with third parties.	enhancing technologies (differential privacy, federated learning); legal or compliance background preferred	
6. AI Risk & Compliance Analyst	Conducts AI risk assessments aligned with NIST AI RMF. Maps regulatory requirements to specific AI products. Maintains the AI risk register. Prepares compliance documentation for audits. Monitors evolving AI regulations and updates internal policies accordingly.	3+ years in risk management or regulatory compliance; knowledge of NIST AI RMF, EU AI Act, sector-specific regulations (FDA, CFPB); analytical and documentation skills; experience with GRC (governance, risk, compliance) platforms; ability to translate regulatory language into actionable requirements	RAI Program Manager (dotted line to Legal/Compliance)
7. AI Ethics Board Member (External Advisor)	Provides independent ethical oversight and challenge to internal decisions. Brings external perspectives from academia, civil society, or affected communities. Reviews high-risk AI deployments before launch. Participates in quarterly Ethics Board meetings. Advises on emerging ethical challenges.	Academic expertise in AI ethics, philosophy of technology, or related disciplines; OR lived experience representing communities affected by AI systems; independence from NovaMind business interests; ability to provide constructive challenge; thought leadership in responsible technology	Reports findings to AI Ethics Board (advisory capacity)

Responsible AI Organizational Structure

The following chart describes how responsible AI governance is organized within NovaMind, showing lines of accountability and decision-making authority. This structure ensures that RAI oversight is both centralized (for consistency) and embedded within product teams (for practical effectiveness).



Accountability Structure

Decision Type	Decision Maker	Consulted	Informed
High-risk AI deployment approval	AI Ethics Board + CAIO	Product team, Legal, Privacy Lead	CEO, Board
Medium-risk AI deployment approval	RAI Program Manager	Fairness Specialist, Compliance Analyst	CAIO, Product leadership
Low-risk AI deployment approval	Product team (self-assessment)	RAI Program Manager (spot-check)	RAI team (logged)
AI incident response escalation	CAIO (with Legal)	AI Ethics Board, affected product team	CEO, Board, affected stakeholders
RAI policy changes	CAIO + AI Ethics Board	All RAI team members, Legal	All product teams, executive leadership



RAI Processes for Vetting AI Products

NovaMind employs distinct but aligned vetting processes for internally-developed AI products and externally-procured AI tools/services. Both processes ensure that AI systems meet the company's ethical standards before reaching production, as recommended by the NIST AI RMF "GOVERN" function and the SCU Ethical Toolkit's emphasis on systematic ethical evaluation.

Process A: Internally-Developed AI Products

1. Pre-Development AI Impact

Assessment: Before any coding begins, the product team completes an AI Impact Assessment documenting intended use, affected populations, risk classification, and explainability requirements. Submitted to RAI Program Manager for review.

2. Design Review Gate: RAI team reviews system architecture, data sources, and model selection for ethical risks. High-risk products require Ethics Board approval to proceed. Fairness requirements and testing criteria defined at this stage.

3. Development-Phase Checkpoints: Embedded RAI specialists participate in sprint reviews. Bias testing begins with training data validation. Explainability methods integrated during model development, not retrofitted. Privacy reviews for data handling.

Process B: Externally-Procured AI Products

1. Vendor RAI Assessment

Questionnaire: Before procurement, vendors must complete NovaMind's RAI questionnaire covering: their own AI ethics policies, bias testing practices, explainability capabilities, data governance standards, incident response procedures, and willingness to share Model Cards or equivalent documentation.

2. Independent Testing & Validation: NovaMind's RAI team independently tests the procured AI product for bias, fairness, and transparency using the company's own test datasets and evaluation criteria. Vendor claims are verified, not taken at face value.

3. Contractual RAI Requirements: Procurement contracts include: mandatory notification of model updates, audit rights, transparency reporting obligations, data processing agreements, liability allocation for AI

4. Pre-Deployment RAI Review:

Comprehensive evaluation including: bias audit results, explainability validation, privacy compliance check, safety testing outcomes, and documentation completeness (Model Card, Data Card, Transparency Notice). Must pass all criteria for deployment authorization.

5. Post-Deployment Monitoring

Activation: Continuous monitoring dashboard activated. Feedback channels established. Quarterly re-evaluation scheduled. Incident response procedures confirmed.

harms, and right to terminate for ethical violations.

4. Integration Review: Before integrating external AI into NovaMind products, review how the external component interacts with internal systems. Ensure end-to-end explainability is maintained. Verify that NovaMind can still provide required explanations to end users.

5. Ongoing Vendor Monitoring:

Annual vendor RAI re-assessment. Monitoring of vendor's public AI ethics track record. Re-testing after vendor model updates. Escalation process if vendor product fails ongoing compliance checks.

Framework Alignment: These processes implement the NIST AI RMF's GOVERN function (GV-1 through GV-6) which emphasizes organizational policies for AI risk management, defined roles and responsibilities, and systematic processes that scale with risk level. The vendor assessment process also aligns with SCU's Ethical Toolkit recommendation to evaluate the full supply chain of AI systems.

12-Month Implementation Timeline

The following timeline outlines NovaMind's plan for establishing the RAI governance organization over 12 months, progressing from foundational setup through full operational maturity:

Month 1–2	Foundation & Leadership: Hire Chief AI Ethics Officer. Secure executive sponsorship and Board mandate for RAI program. Allocate initial budget (\$500K Year 1). Draft RAI charter defining scope, authority, and objectives. Engage external advisory candidates.
Month 2–3	Team Building: Hire RAI Program Manager and AI Risk & Compliance Analyst. Begin recruitment for Fairness Specialist and Transparency Engineer. Formalize AI Ethics Board membership (3 internal + 2 external advisors). Establish meeting cadence (monthly initially).
Month 3–4	Policy Development: Publish NovaMind AI Ethical Principles (ratified by Board). Develop AI Impact Assessment template and risk classification framework. Create RAI review process documentation. Define roles and RACI matrix for AI governance decisions.
Month 4–5	Tooling & Infrastructure: Hire Privacy & Data Governance Lead. Deploy bias testing tools (Fairlearn, AI Fairness 360) into ML pipeline. Implement explainability libraries (SHAP, LIME) in development environments. Set up AI risk register and documentation repository.
Month 5–6	Training & Awareness: Deliver mandatory RAI training to all 150+ AI/ML engineers (covering bias, explainability, privacy, safety). Train product managers on AI Impact Assessments. Brief executive team on governance structure and escalation procedures. Publish internal RAI knowledge base.
Month 6–7	Pilot Application: Apply full RAI review process to one high-risk product (Clinical Decision Support System) as pilot. Conduct complete bias audit, explainability validation, and documentation review. Identify process gaps and refine procedures based on pilot learnings.

Month 7–8

Expansion to All High-Risk Products: Extend RAI vetting process to Credit Risk Scoring Engine. Begin vendor assessment process for any externally-procured AI components. Implement pre-deployment gates in product release workflows. Activate continuous monitoring dashboards.

Month 8–9

Medium-Risk Coverage: Apply RAI processes to Customer Service Chatbot. Deploy user-facing AI disclosure features. Establish external feedback/complaint channel for AI-related concerns. Refine KPI tracking and reporting mechanisms.

Month 9–10

Full Coverage & Integration: Extend baseline RAI self-assessment to low-risk products (Document Summarization). Integrate RAI checkpoints into standard agile development workflow. All new AI projects now follow RAI process from inception. Complete vendor RAI assessments for all external AI tools.

Month 10–11

Audit & Assessment: Conduct internal RAI maturity assessment. Prepare for external third-party audit. Address identified gaps. Compile first annual RAI report with metrics, incidents, and improvement actions. Update policies based on 10 months of operational experience.

Month 11–12

Maturity & Sustainability: Complete first third-party RAI audit. Publish external transparency report (annual commitment). Set Year 2 goals and budget. Transition AI Ethics Board to quarterly cadence. Establish continuous improvement cycle. Plan for emerging challenges (generative AI governance, multimodal AI oversight).

Key Performance Indicators (KPIs) for Responsible AI

Following the KPI best practices from *Ethical Machines* (Chapter 6), NovaMind's RAI KPIs are designed to be **specific, measurable, actionable, time-bound, and tied to accountability**. Each KPI has a clear owner, target, measurement method, and escalation trigger to ensure they drive real behavior change rather than serving as passive metrics.

KPI	Ethical Principle Aligned	Target	Measurement Method	Owner	Escalation Trigger
Bias Detection Rate – Percentage of AI models that complete full bias audit before deployment	Fairness & Non-Discrimination	100% of high/medium-risk models audited before production release	Automated tracking in CI/CD pipeline; bias audit completion logged in AI governance platform	AI Fairness & Bias Specialist	Any high-risk model deployed without completed bias audit triggers immediate CAIO review and potential rollback
Fairness Metric Compliance – Percentage of models meeting defined fairness thresholds across protected groups	Fairness & Non-Discrimination	≥95% of deployed models meet equalized odds threshold (ratio within 0.8–1.25 across groups)	Automated fairness testing in production monitoring; disaggregated performance reports by demographic subgroup	AI Fairness & Bias Specialist	Any model falling below threshold for >7 days triggers mandatory remediation plan within 48 hours
Explainability Coverage – Percentage of	Transparency & Explainability	100% of high-risk decisions (clinical,	System logging of explanation generation;	AI Transparency &	Coverage dropping below target

KPI	Ethical Principle Aligned	Target	Measurement Method	Owner	Escalation Trigger
high-risk AI decisions that include user-accessible explanations		credit) include explanation; 80%+ for medium-risk	user interface audit; sample verification of explanation accuracy	Explainability Engineer	for >2 weeks triggers engineering sprint to restore compliance
User Comprehension Score – Percentage of users who correctly understand AI explanations in periodic testing	Transparency & Explainability	≥75% of tested users demonstrate correct understanding of how AI reached its output	Quarterly user comprehension surveys with representative stakeholder samples; task-based comprehension tests	RAI Program Manager + UX Research	Score below 60% triggers explanation redesign initiative within 30 days
AI Impact Assessment Completion – Percentage of new AI projects that complete Impact Assessment before development begins	Accountability & Governance	100% of new AI projects have approved Impact Assessment before sprint 1	Project management system tracking; RAI team sign-off records; audit trail in documentation repository	RAI Program Manager	Any project starting development without approved AIA triggers immediate stop-work and CAIO notification
RAI Training Completion – Percentage of AI/ML team members who complete annual RAI training	Accountability & Governance	100% completion within 30 days of onboarding (new hires) and annually (existing staff)	LMS (Learning Management System) completion tracking; manager attestation	RAI Program Manager + HR	Non-completion after 45 days triggers manager escalation; after 60 days

KPI	Ethical Principle Aligned	Target	Measurement Method	Owner	Escalation Trigger
					restricts production deployment access
Incident Response Time – Time from AI ethics incident detection to initial response action	Safety & Reliability	High-severity: response within 4 hours; Medium: within 24 hours; Low: within 5 business days	Incident management system timestamps; time from ticket creation to first action logged	CAIO + RAI Program Manager	Any high-severity incident exceeding 4-hour response window triggers automatic Board notification
Data Privacy Compliance Rate – Percentage of AI systems passing quarterly privacy audit	Privacy & Data Governance	100% of production AI systems pass quarterly privacy review with zero critical findings	Quarterly privacy audits; automated data handling checks; consent management system reporting	AI Privacy & Data Governance Lead	Critical privacy finding triggers 72-hour remediation deadline and potential system suspension
Human Override Availability – Percentage of time human review/appeal options are accessible for	Human Autonomy & Oversight	99.5% availability of human review pathways for all high-risk AI decisions	System uptime monitoring for human-in-the-loop interfaces; appeal request processing time tracking	Product Engineering + RAI Program Manager	Availability dropping below 99% for >24 hours triggers incident response

KPI	Ethical Principle Aligned	Target	Measurement Method	Owner	Escalation Trigger
high-risk AI decisions					and potential AI system pause
Vendor RAI Compliance – Percentage of external AI vendors meeting NovaMind's RAI standards	Accountability & Governance	100% of active AI vendors pass annual RAI assessment; new vendors assessed before contract signing	Vendor assessment questionnaire scores; independent testing results; contract compliance reviews	AI Risk & Compliance Analyst + Procurement	Vendor failing assessment triggers 90-day remediation plan; continued failure initiates vendor replacement

KPI Design Principles (from *Ethical Machines*): Each KPI is (1) tied to a specific ethical principle, (2) has a concrete measurable target, (3) assigns clear ownership, (4) includes an escalation trigger that makes inaction impossible, and (5) is reviewed quarterly by the AI Ethics Board to assess whether targets remain appropriate and whether KPIs are driving genuine ethical improvement rather than checkbox compliance.

Summary & Conclusion

Governance Framework Summary

- **7 dedicated RAI roles** providing specialized expertise in ethics leadership, program management, fairness, transparency, privacy, compliance, and external oversight
- **Clear organizational structure** with defined accountability from Board level through product teams, ensuring both top-down commitment and ground-level implementation
- **Dual vetting processes** for internal and external AI products, ensuring consistent ethical standards regardless of whether AI is built or bought
- **12-month implementation timeline** moving from foundational hiring and policy through pilot application to full operational maturity with third-party validation
- **10 actionable KPIs** aligned with ethical principles, each with clear targets, measurement methods, ownership, and escalation triggers to drive accountability

Key Design Decisions

- Centralized RAI team for consistency
+ embedded specialists in product

Personal Reflections

Building a governance framework that is both comprehensive and practical is a significant challenge. The tension between being thorough (to catch all potential ethical issues) and being efficient (so teams can actually ship products) was ever-present in designing these processes. An overly burdensome process will be circumvented; an overly light one will miss critical issues.

The concept from *Ethical Machines* that resonated most strongly was the emphasis on making KPIs truly actionable through escalation triggers. Without consequences for non-compliance, even well-designed governance becomes advisory rather than authoritative.

The NIST AI RMF provided a useful structural framework but required significant interpretation to apply to a specific company context. The challenge of translating general principles into concrete organizational roles, processes, and metrics cannot be underestimated.

teams for practical impact

- Risk-proportionate processes: high-risk AI faces rigorous review; low-risk follows streamlined self-assessment
- External advisors on Ethics Board to prevent echo chambers and ensure community perspectives
- KPIs designed with escalation triggers (per *Ethical Machines*) to ensure metrics drive action, not just reporting

References: Reid Blackman, *Ethical Machines: Your Concise Guide to Totally Unbiased, Transparent, and Respectful AI* (Harvard Business Review Press, 2022), Ch. 6; NIST AI Risk Management Framework (AI RMF 1.0), January 2023; Markkula Center for Applied Ethics, "An Ethical Toolkit for Engineering/Design Practice," Santa Clara University; EU AI Act (2024); Google AI Principles (2018); Microsoft Responsible AI Standard v2 (2022); IBM Everyday Ethics for AI (2019).

